Research Seminar

Group 8

11. FLOW TOXICITY AND LIQUIDITY IN A HIGH-FREQUENCY WORLD

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Review of Financial Studies 25(5), 2012

Nazarov · Doroshenko · Burlakov · Daunov · Sergeev

Why toxic order flow matters

- **Market makers** (today mostly high-frequency firms) earn a tiny spread on many trades and avoid directional bets.
- They lose when the other side **knows more**. Order flow is **toxic** when it adversely selects them, often before they notice.
- When toxicity rises they widen quotes or **leave**. With no one absorbing orders, prices gap. The paper links the **6 May 2010 flash crash** to this.
- **Question:** can toxicity be measured in real time, from trades alone, early enough to act?



Method: from PIN to VPIN, measured in volume time

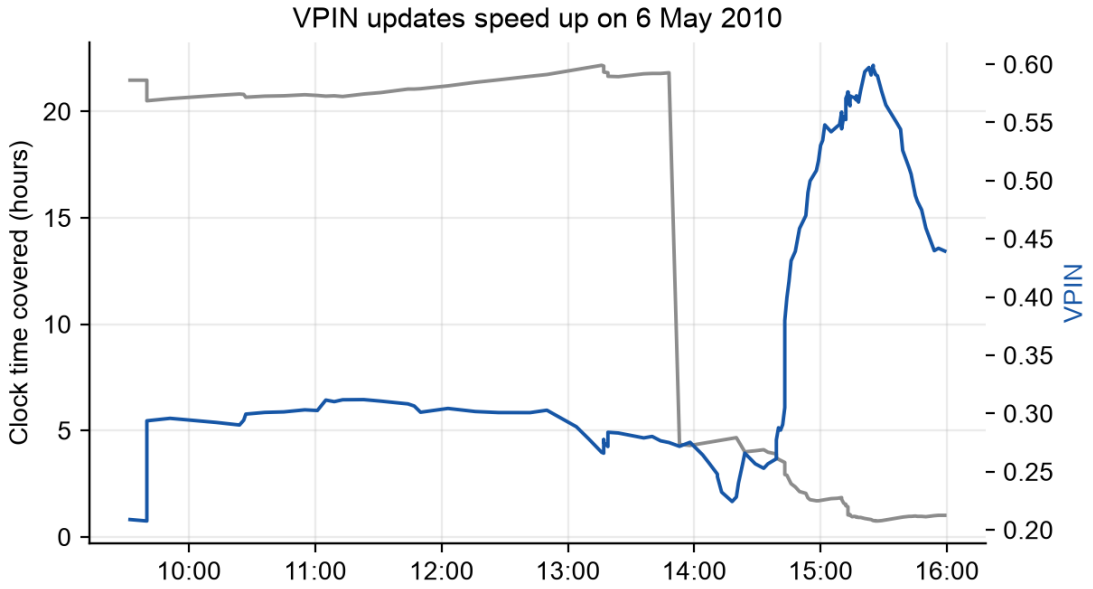
• **PIN** (Easley et al. 1996): the share of orders that come from informed traders

$$\mathrm{PIN} = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon}$$

• α: chance of news, μ: informed arrivals, ε: uninformed arrivals per side. Needs maximum likelihood: slow and unstable at high frequency.

• **VPIN:** |buys − sells| estimates αμ, total volume estimates αμ + 2ε. No parameters to fit.

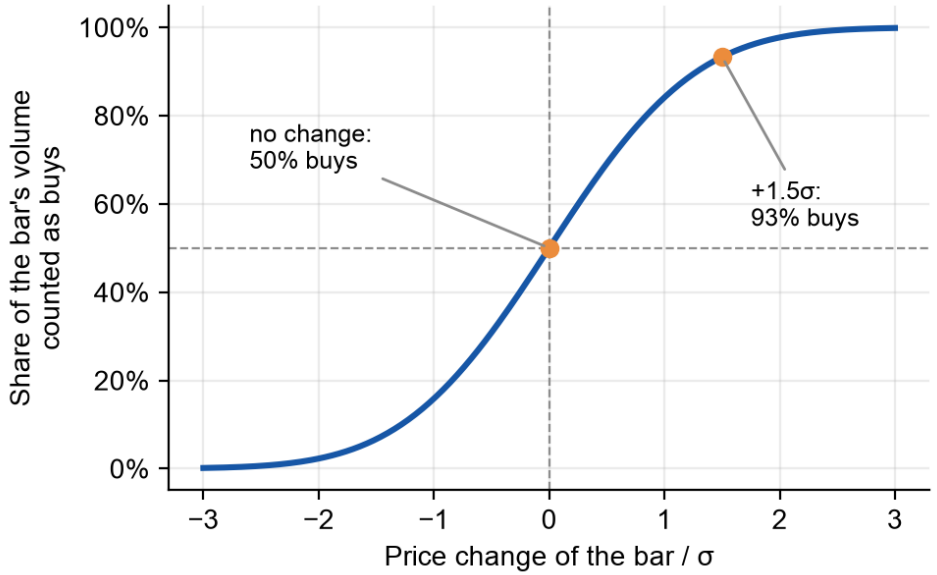
• **Volume clock:** cut trading into buckets of equal volume (1/50 of a day). Busy, news-heavy periods get more updates.



Method: bulk volume classification

$$V_\tau^B = \sum_{i \in \tau} V_i Z\left(\frac{P_i - P_{i-1}}{\sigma_{\Delta P}}\right), \quad V_\tau^S = V - V_\tau^B$$

- Group trades into **one-minute bars**; split each bar's volume by its standardized price change.
- No price change → 50/50. A rise of 1.5σ → 93% buys. Z is the standard normal CDF; σ is the standard deviation of bar price changes.
- Robust to order splitting and cancellations, cheap to compute.
- **But:** the imbalance it produces grows with |price change|, by construction.



Data, and what is estimated, chosen and evaluated

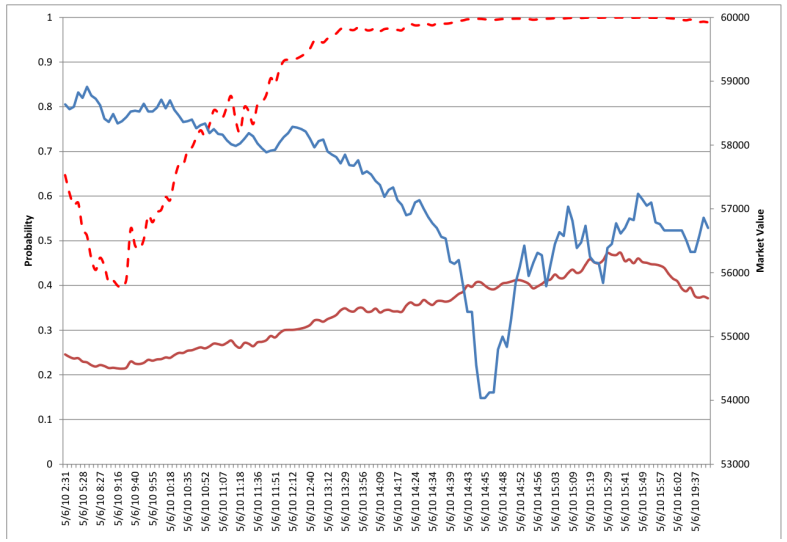
Data	Paper	Ours
Market and hours	E-mini S&P 500 and WTI crude futures; almost 24 hours	IVE (S&P 500 Value ETF) ticks with quotes, 09:30–16:00 New York; BTCUSDT for extensions
Period	Jan 2008 – Aug 2011 (incl. 2008 crisis)	2009-09-28 – 2011-08-15; out of sample to Sep 2026
Classification, settings	bulk, one-minute bars; V = ADV/50, n = 50, whole-sample σ and CDF	identical, plus 10-second bars and Lee-Ready

Step	Objective	Data used
Estimate PIN (original model)	Maximise the likelihood of a Poisson mixture: α, μ, ε, δ	daily buy and sell counts
Estimate VPIN	Nothing is fitted: average |sell − buy volume| / V, a moment estimate of αμ / (αμ + 2ε)	the last n buckets
Inputs σ and V	Standard deviation of one-minute price changes; average daily volume / 50	paper: whole sample · ours, real-time: previous 250 days
Choose buckets/day and n	Paper: grid cell with the highest corr(ln VPIN, next |return|) (Fig. 10) or true-positive rate (Table 5)	paper: the evaluation sample itself · ours: 2009–11, then frozen
Evaluate	Corr with next |return|; P(VPIN high | big move); true positives. We add lift and ΔR^2 over volatility	paper: in sample · ours: 2011–26, out of sample

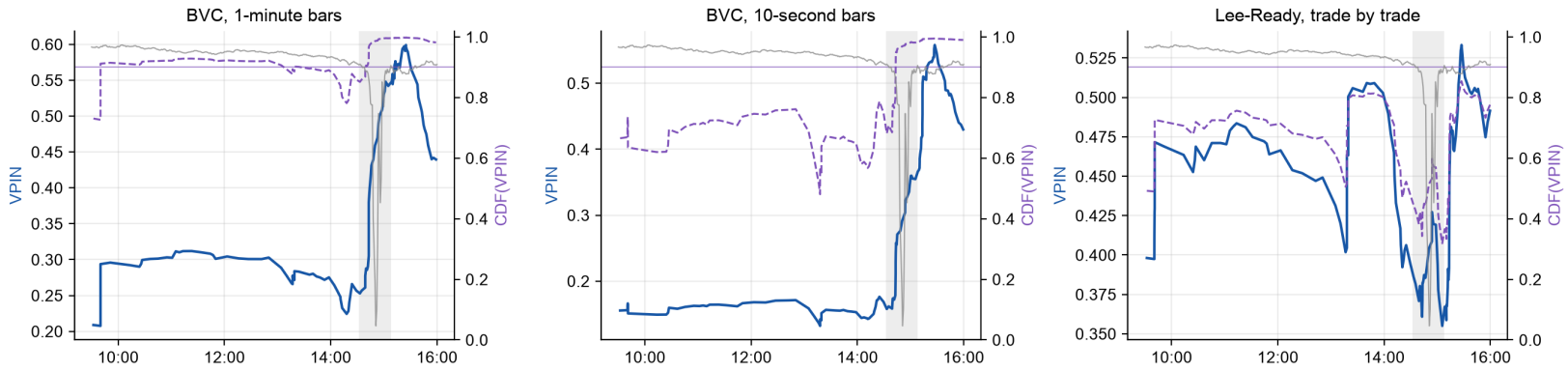
• Also removed: 42 IVE trades at prices like $0.08 that the exchanges **cancelled** on 6 May 2010. 29% of IVE one-minute bars have no price change; bulk classification splits them 50/50. ± on our numbers: block bootstrap over calendar months (1,000 replications), because VPIN windows overlap and volatility clusters.

• **VPIN has no training objective.** The paper's only optimisation is the parameter search, and it maximises the statistic it then reports, on the same data. We select on 2009–11 and evaluate on 2011–26.

Result 1: the flash crash, paper vs ours



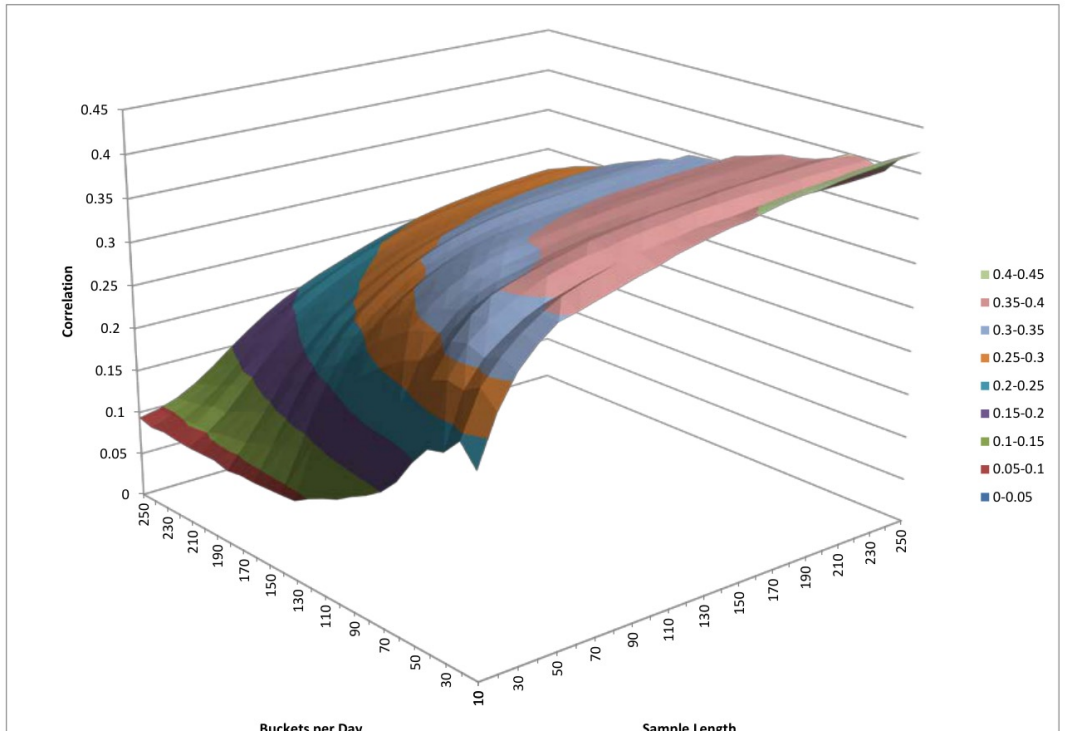
Paper, Fig. 4(a): E-mini, one-minute bars



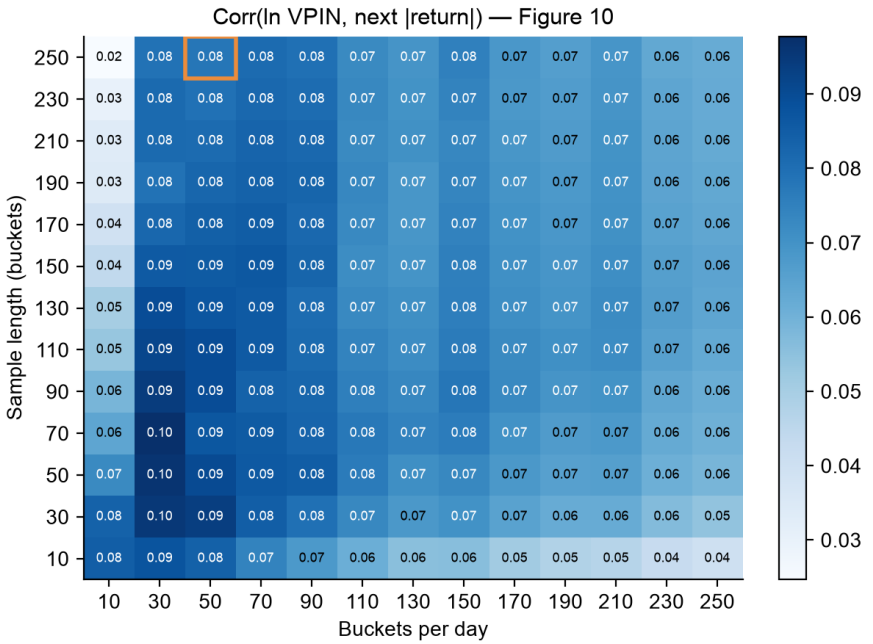
Ours: IVE, 6 May 2010, three classification schemes

- **One-minute bars:** CDF > 0.9 from 09:40, **4.9 h** before the fall, as the paper says.
- **10-second bars:** first above 0.9 at 14:43, during the fall. **Lee-Ready:** never (max CDF 0.85).
- One event: no ± possible. The warning depends on the classification choice.

Result 2: does VPIN predict the next bucket's volatility?



Paper (E-mini): 0.40 at (50, 250), 95% CI [0.392, 0.408] (iid)



Ours (IVE): 0.08 ± 0.05 at (50, 250), 95% CI [0.04, 0.22]

On IVE

Same sign; 0.40 is outside our interval.

About the ±

An iid interval would give ours as [0.07, 0.09]: overlapping windows make it far too narrow.

On every window

None of 41 two-year windows (IVE, BTC) reaches the paper's 0.40, nor its > 84% for big moves; we could detect ≥ 0.13.

Reproduction scorecard

Paper reports: E-mini, 2008–11

#	Claim	Reported
1	VPIN does not depend on where the bucket sequence starts (std / mean across 1,000 starts)	6.5%
2	Volume time is closer to normal than clock time: Kurt, J-B*, L-B*, White* lower	4 of 4 at 50 and at 100 draws a day
3	CDF(VPIN) > 0.9 at least 2 h before the flash crash: BVC, one-minute bars	≥ 2 h
4	... BVC, 10-second bars	≥ 2 h
5	... Lee-Ready, trade by trade	≥ 2 h
6	Low VPIN (bottom quartile) → next |return| < 0.25%	> 90%
7	Corr(ln VPIN, next |return|) at (50, 250)	0.40 [0.392, 0.408]
8	P(VPIN in top quartile | next |return| > 0.75%) at (50, 250)	> 84%
9	P(move > 0.75% while VPIN stays in the top quartile) at (50, 250)	51.8%
10	... at (10, 350)	78.6%

Ours: IVE, 2009–11

#	Ours (± s.e.)	95% interval	Verdict
1	3.7%	–	reproduced
2	1 of 4 / 2 of 4 significant	P ≥ 0.95	partly
3	4.9 h	–	reproduced
4	only during the fall	–	not reproduced
5	never (max CDF 0.85)	–	not reproduced
6	96.1% ± 0.5%	[95.0%, 97.2%]	reproduced
7	0.08 ± 0.05	[0.04, 0.22]	weaker
8	51% ± 13%	[19%, 71%]	weaker
9	34% ± 4%	[26%, 43%]	weaker
10	60% ± 12%	[37%, 85%]	inconclusive

How could these results be overfit?

In the paper

• The CDF of VPIN is built from the **whole sample**, future included.

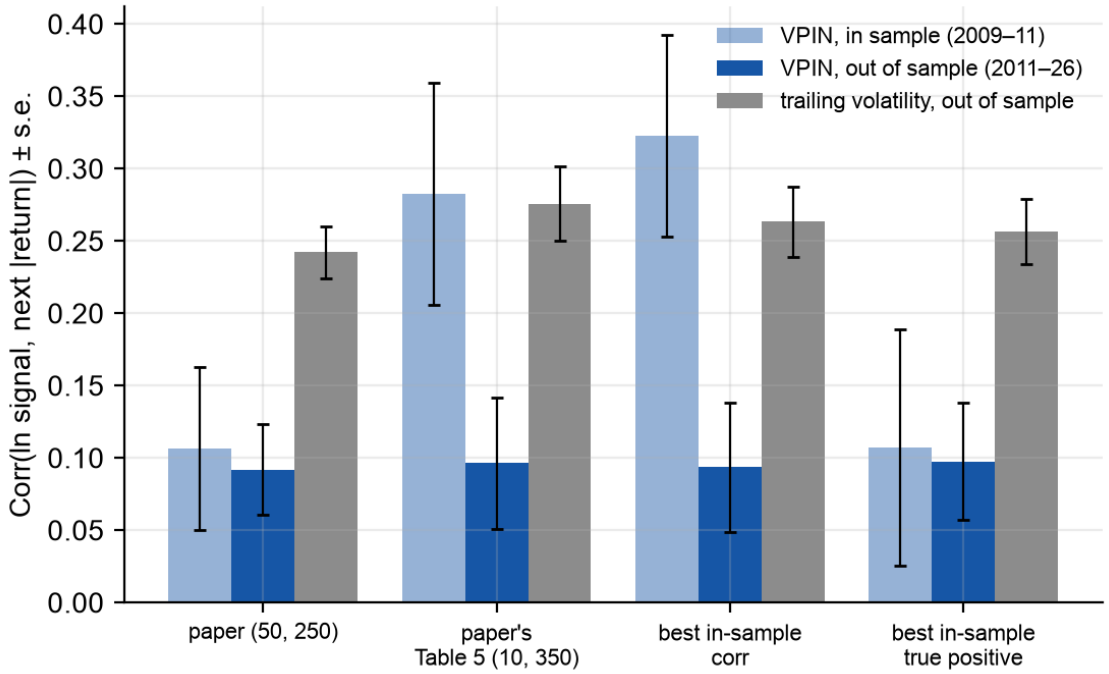
• Bucket size and window are picked by **scanning grids** on the same data (Figs. 10, 12, 14); Table 5 reports the best cell.

• **No out-of-sample** period; its ± assumes independent buckets. One flash crash = one observation.

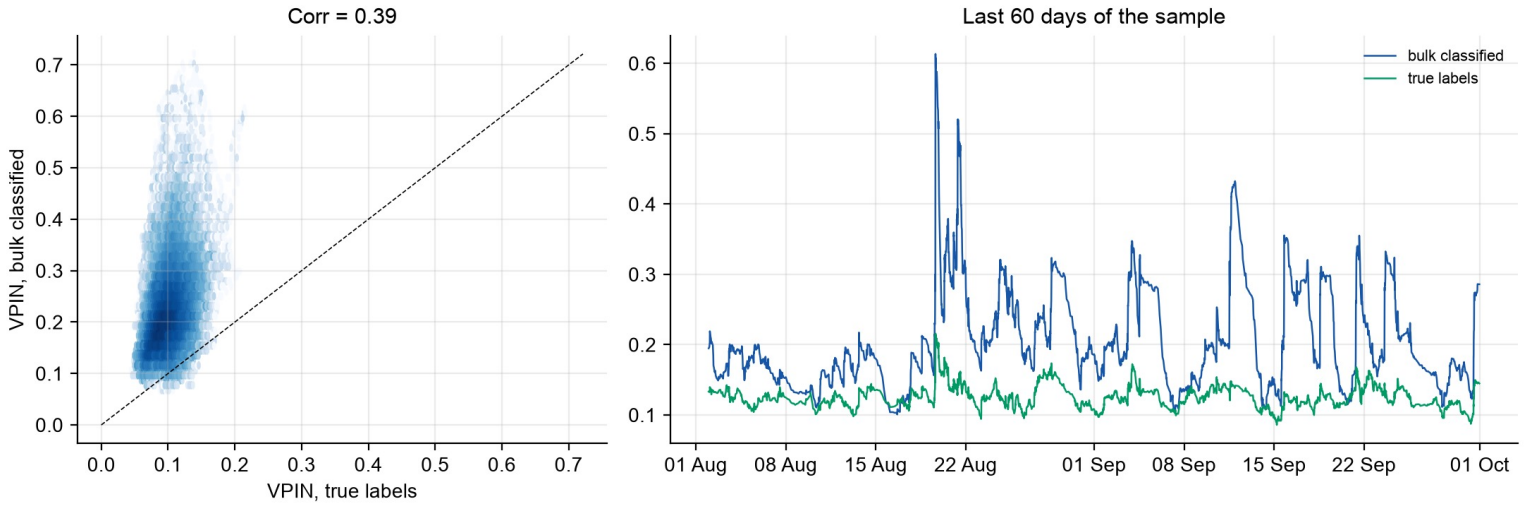
Our check

• Real-time VPIN: σ, V and CDF from **past data only**; parameters frozen in 2009–11, tested on 2011–26.

• Best in-sample true-positive pair (20, 500): **92% ± 5%** → **45% ± 3%** out of sample (bottom quartile: 47% ± 4%).



Extension 1: is bulk classification right? A ground-truth test



- Bars: BVC gets the sign right for **82.1% ± 0.2%** of volume.
- Buckets: BVC inflates |imbalance| **2.2×**; the two VPINs correlate only **0.39 ± 0.04**.

On 5 days with every trade

- Tick rule: **93%–97%** of volume classified right.
- Bucket imbalance vs truth: tick rule **≥ 0.99**, BVC **0.51–0.74** (1-min bars).
- Ranges are across the 5 days. Caveat: Binance aggregates the fills of one order, which helps the tick rule.
- **The paper's claim that bulk beats trade-by-trade classification does not hold here.**

Extension 2: is VPIN more than volatility?

BTCUSDT 2020–26, 50 buckets/day, n = 50	Corr with next |return|	P(top quartile | move > 0.75%)	Time in top quartile	Lift
VPIN (bulk classified)	0.13 ± 0.02	41% ± 3%	27%	1.53 ± 0.08
VPIN (true labels)	-0.12 ± 0.02	17% ± 3%	29%	0.61 ± 0.08
RV (trailing volatility)	0.36 ± 0.03	40% ± 7%	12%	3.31 ± 0.28

- VPIN from **true labels** is negatively related to the next move; bulk-classified VPIN is positive; **trailing volatility wins**.
- BVC imbalance moves with the bucket's own |return| (0.44 ± 0.01 vs 0.29 ± 0.02 for the truth): BVC builds volatility into VPIN.
- Regression with volatility and trade-intensity controls: VPIN adds ΔR^2 = **0.0010 ± 0.0005** on BTC, **0.0025 ± 0.0011** on IVE out of sample.

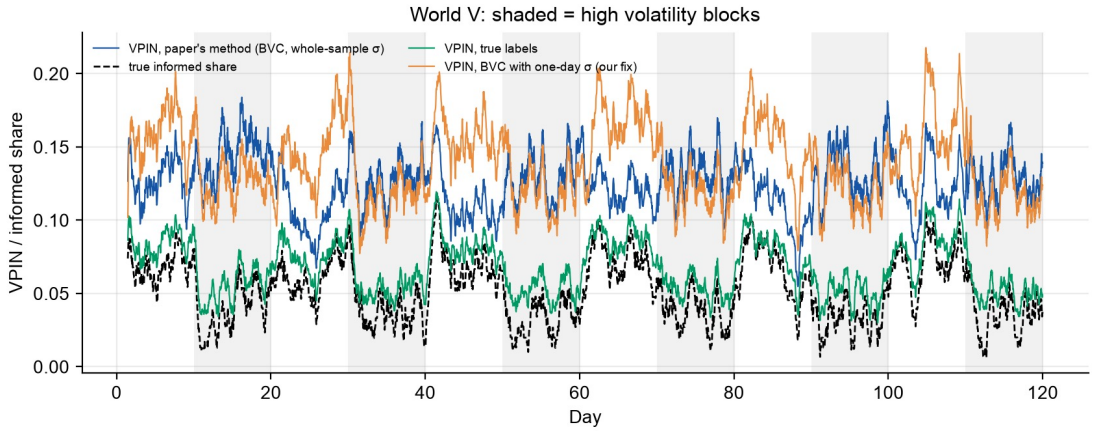
Extension 3: a known-answer test of VPIN (pre-registered)

- **Gap in the paper:** its Monte Carlo (Appendix A.2) uses correctly labelled trades and no prices; its VPIN uses bulk classification, which needs prices.
- **Our model:** the paper's PIN model, plus a price: $\Delta p = \lambda$ (buys − sells) + public noise. Informed flow moves the price; public noise does not involve any trader.
- **Worlds:** T = more informed trading in some blocks; V = more volatility and uninformed volume, informed trading unchanged; TV = both; 0 = neither.
- **Same pipeline**, truth known, 200 replications per world, rules fixed before running.

Rule	Pre-registered statement	Result	Verdict
D1	true labels track informed trading	0.98	holds
D2	paper's VPIN: false alarms from volatility	+0.118 ± 0.002	holds
D3	paper's predictive test fooled by volatility	-0.007 ± 0.001	does not hold
D4	one-day σ: fewer false alarms, same detection	-0.301; AUC 0.992	holds

Tracking of the true informed share, world V

- paper's VPIN **0.03** · true labels **0.96** · one-day σ (fix) **0.83**
- **When volatility changes, the paper's VPIN stops measuring informed trading.**



World V: the paper's VPIN (blue) stays high in volatile blocks while the true informed share (dashed) falls.

Extension 4: the fix on real data

Market (real-time, 50 buckets/day)	Signal	Corr with next |return|	Lift
BTC	VPIN, standard BVC	0.127 ± 0.020	1.53 ± 0.08
BTC	**VPIN, local-sigma BVC (fix)**	0.051 ± 0.017	1.27 ± 0.07
BTC	trailing volatility	0.359 ± 0.032	3.31 ± 0.28
BTC	VPIN, true labels	-0.123 ± 0.024	0.61 ± 0.08
IVE 2011-26	VPIN, standard BVC	0.092 ± 0.031	1.57 ± 0.20
IVE 2011-26	**VPIN, local-sigma BVC (fix)**	0.007 ± 0.023	1.30 ± 0.24
IVE 2011-26	trailing volatility	0.242 ± 0.018	2.76 ± 0.28

Closer to the truth (BTC)

• corr with true-label VPIN: fix **0.47** vs standard 0.39 (+0.08 ± 0.02)

Less volatility in disguise

• its correlation with the next move collapses: the paper's predictive power came from the bias

What remains beyond volatility

• IVE: t = 8.3, $\Delta R^2 = 0.0029$; BTC: negative (t = -4.3)

What our work adds to the paper

• **1. Closes the paper's validation gap.** Its Monte Carlo never runs bulk classification. Our known-answer simulation of its own model shows that VPIN, as the paper computes it, stops measuring informed trading when volatility changes.

• **2. Locates where VPIN's predictive power comes from.** With true labels VPIN does not predict volatility (BTC); with shuffled directions it predicts as well as the real VPIN (IVE, BTC). It is the size of price moves, not toxicity.

• **3. A fix that works where it can be checked.** Measuring price changes against the previous day's volatility makes VPIN track true toxicity better on BTC, and removes the volatility artefact.

• **4. An honest evaluation.** Real-time, out of sample 2011–2026, all 41 two-year windows, pre-registration, ± on every metric, and a ledger with multiple-testing control. The paper's predictive numbers do not appear anywhere.

Confirms on new data: Andersen & Bondarenko (2014, 2015); Chakrabarty, Pascual & Shkilko (2015).

Placebo: direction of every imbalance shuffled

Sample	Corr, real VPIN	Corr, direction shuffled	Kept
IVE 2009-11, paper's method (50, 250)	0.078	0.078 ± 0.002	101%
IVE 2011-26 (50, 250)	0.092	0.090 ± 0.001	98%
BTC 2020-26 (50, 50)	0.127	0.128 ± 0.001	101%

• **The direction of trading carries almost none of VPIN's predictive power:** the size of price moves does.

• **Leave one year out:** 0 sign flips across 5 signal–sample pairs.

• **Power:** the reproduction detects any correlation ≥ 0.13; the paper's 0.40 would not be missed.

• **Ledger:** 26 tests behind our claims; supported after Benjamini–Hochberg **17**, after Bonferroni **13**; significant but on the wrong side **3** (reported as not supported).

• New tests pre-registered before running; every later change logged.

Critique and next steps

Critique

• Bulk classification **builds volatility into VPIN**; with true labels the predictive link disappears.

• Whole-sample CDF and grid-searched parameters: **look-ahead and selection** with no out-of-sample test.

• Main evidence is **one event** (6 May 2010); on IVE the warning depends on bar size and classifier.

• No baseline: the paper never compares VPIN with **plain volatility**, which wins in our tests.

• Its **±** treats overlapping buckets as independent: far too narrow.

• The authors hold a **patent** on VPIN and disclose a financial interest.

With three more months

• E-mini futures with **exchange aggressor flags** (e.g. CME MDP 3.0): the paper's market, with ground truth.

• VPIN as a **filter for a market-making strategy**: does it reduce adverse-selection losses after costs?

• Correct the grid search for **multiple testing** (White's reality check, deflated statistics).

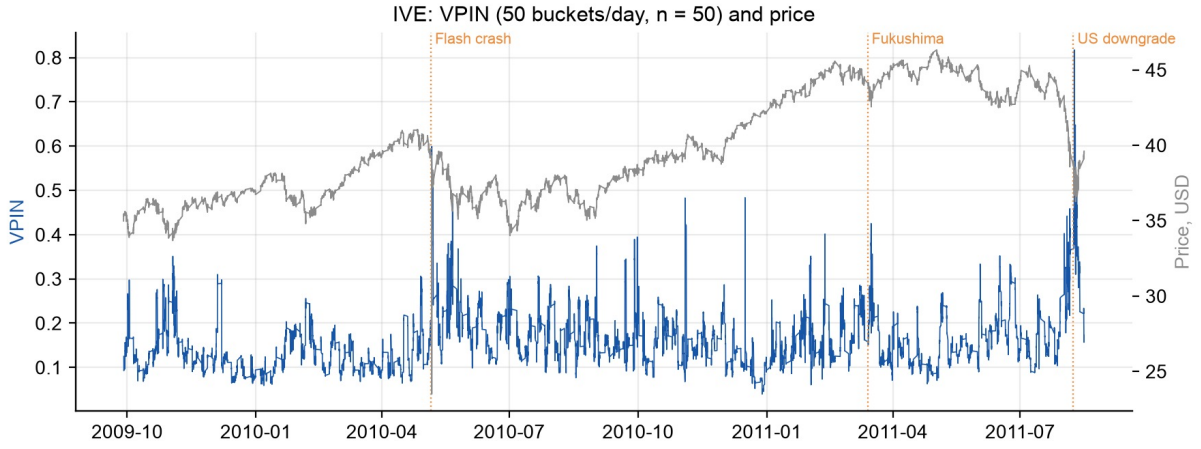
• Combine with **HAR volatility forecasts** and test VPIN's incremental value out of sample.


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Backup: the VPIN metric and how it is read

$$\mathrm{VPIN}_\tau = \frac{\sum_{j=\tau-n+1}^{\tau} \left| V_j^S - V_j^B \right|}{nV}$$

- Bucket size **V = average daily volume / 50**; window **n = 50** buckets ≈ one day.
- Updated after **every bucket**: on busy days many times per hour.
- Levels depend on classification and settings, so the paper reads **CDF(VPIN)** over the whole sample. Alarm: CDF > 0.9.
- Claim: high VPIN warns of **toxicity-induced volatility**.



Backup: the volume clock and the stability of VPIN

Paper: E-mini, 2008–11

50 draws a day	Clock	Volume
Kurt	34.25	**20.52**
J-B*	40.69	**12.82**
L-B*	40.73	**24.84**
White*	0.10	**0.04**
Max	20.83	**12.26**

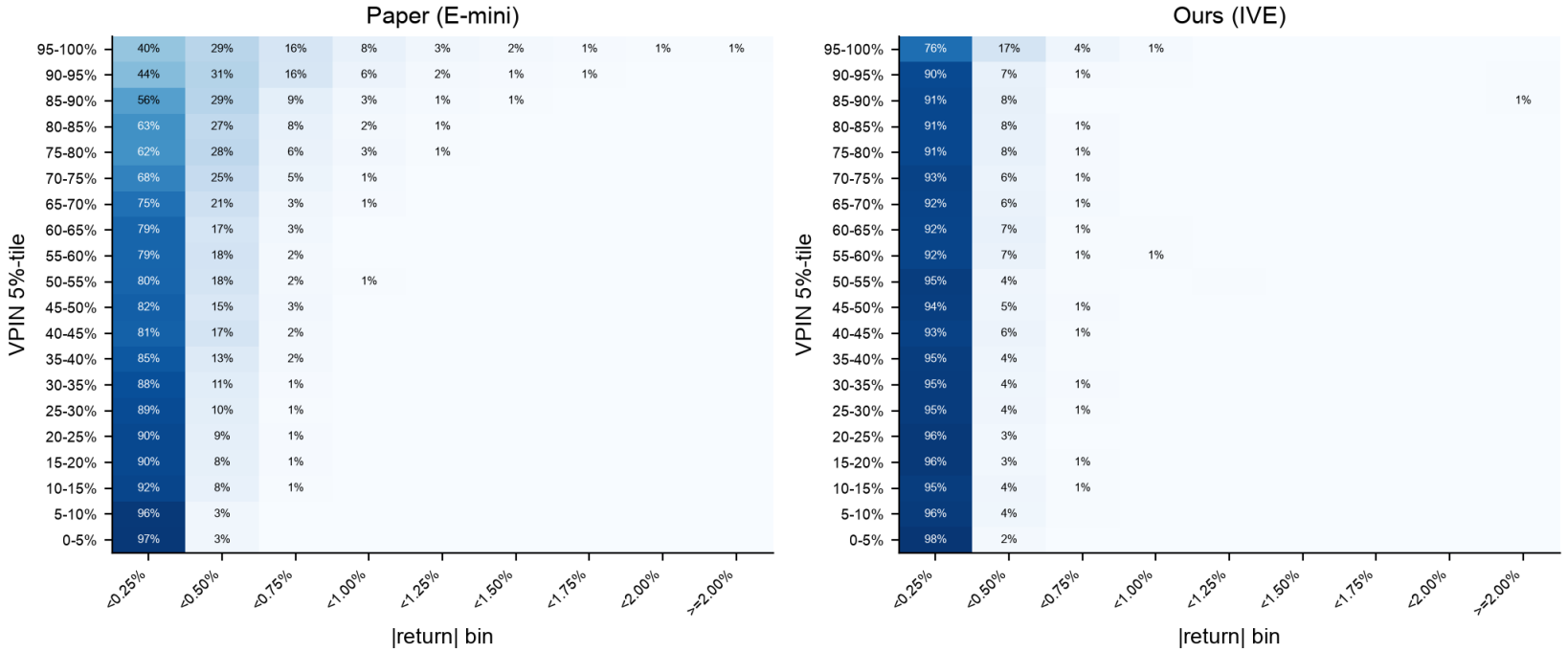
Ours: IVE, 2009–11 (± s.e.)

50 draws a day	Clock	Volume	P(volume lower)
Kurt	13.12 ± 3.96	9.05 ± 1.81	0.947
J-B*	4.27 ± 3.11	1.53 ± 0.89	0.947
L-B*	23.13 ± 32.34	43.03 ± 25.23	0.313
White*	0.28 ± 0.10	0.16 ± 0.06	**0.995**
Max	13.08 ± 2.29	9.86 ± 1.51	**0.977**

- Ours: volume time is **significantly** lower on 1 of the 4 test statistics at 50 draws a day and 2 of 4 at 100; tails and White* clearly improve, autocorrelation (L-B*) does not at 50.
- **Stability (Fig. 5):** restarting the buckets 0, 1, …, 999 bars later changes VPIN by 3.7% of its mean on IVE; the paper reports 6.5% for the E-mini.

Backup: high VPIN and large moves (Tables 4 and 5)

Table 4a: P(next |return| bin | VPIN 5%-tile), (50, 250)



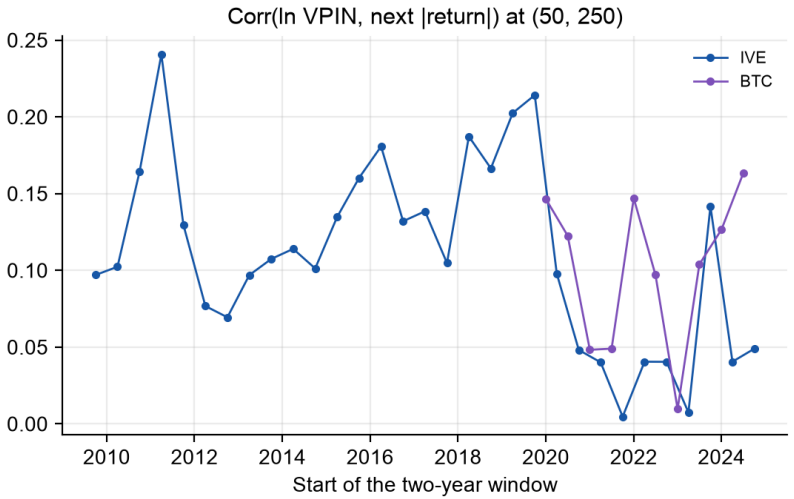
Paper reports (E-mini)

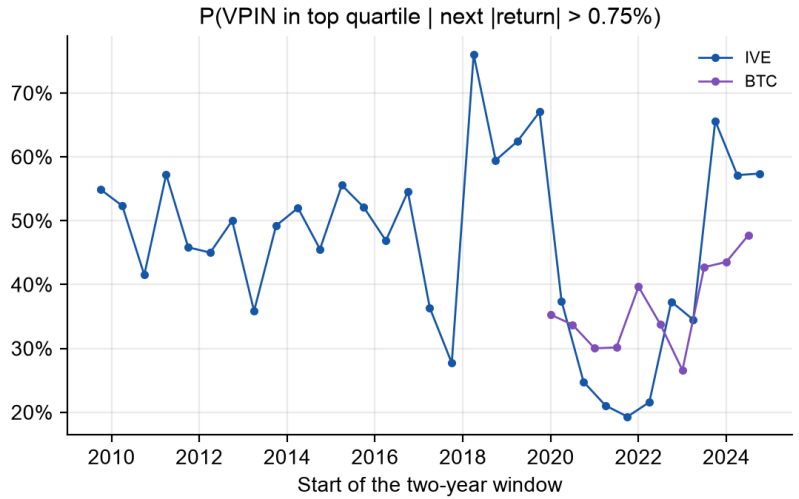
- Bottom-quartile VPIN → next |return| < 0.25%: > 90%
- P(VPIN top quartile | move > 0.75%): > 84%
- Big move while VPIN stays high, (10, 350): 78.6%

On IVE (± s.e.)

- Bottom quartile → < 0.25%: **96.1% ± 0.5%**
- P(top quartile | move > 0.75%): **51% ± 13%**
- Big move while VPIN stays high, (10, 350): **60% ± 12%**, 95% CI [37%, 85%]
- **Low VPIN = calm reproduces; the warning value is weaker or uncertain.**

Backup: the paper's predictive claims on every two-year window





- IVE: corr 0.00 to 0.24; P(top quartile | big move) 19% to 76%.
- BTC: corr 0.01 to 0.16; P(top quartile | big move) 27% to 48%. Big moves are routine on BTC, so low VPIN scores as many true positives.

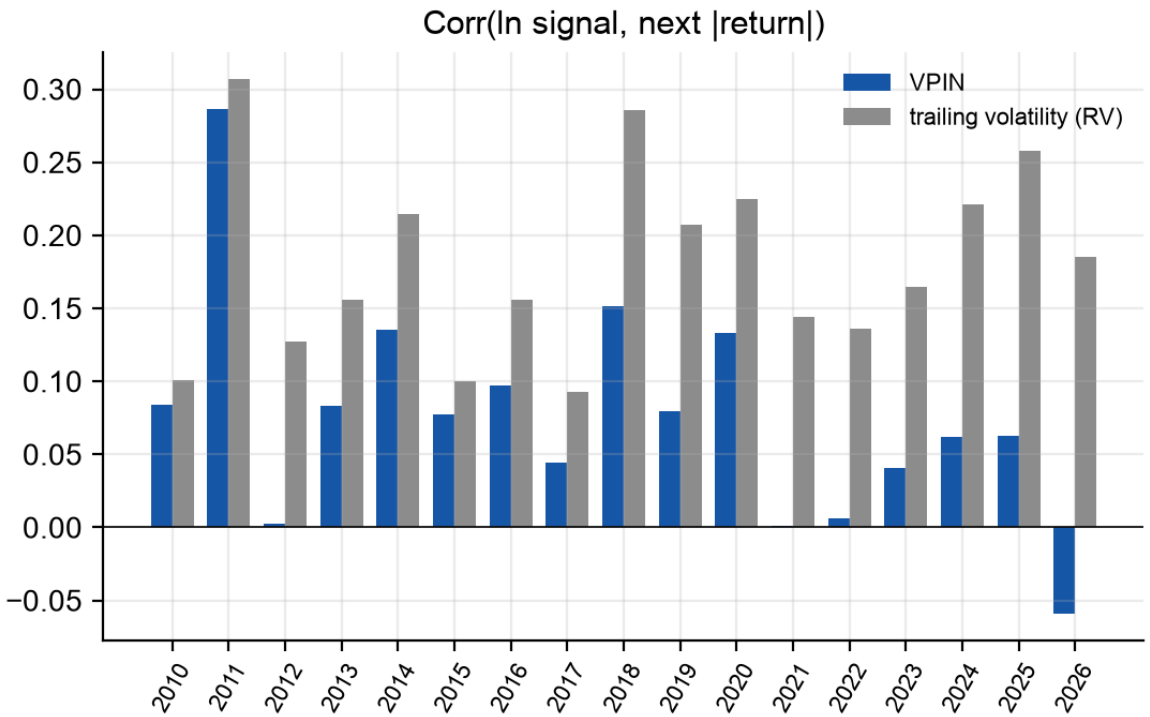
Paper reports (E-mini, 2008–11)

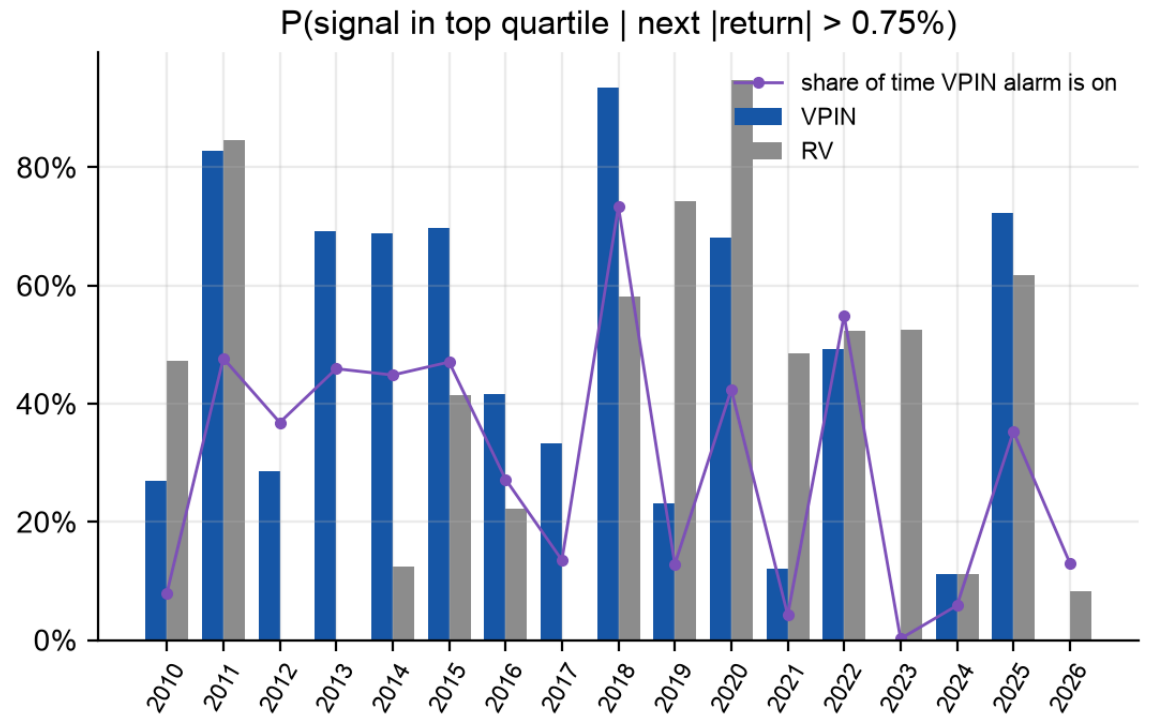
- corr 0.40; P(top quartile | big move) > 84%

Our reading

- No window on either market reaches the paper's numbers.
- Our reproduction could detect any correlation ≥ 0.13 (80% power): the gap is not a matter of power.

Backup: real-time VPIN out of sample, year by year





• Paper's (50, 250), 2011–26: corr VPIN **0.09 ± 0.03** vs trailing volatility **0.24 ± 0.02**; lift **1.57 ± 0.20** vs **2.76 ± 0.28**.

Backup: crashes after the paper

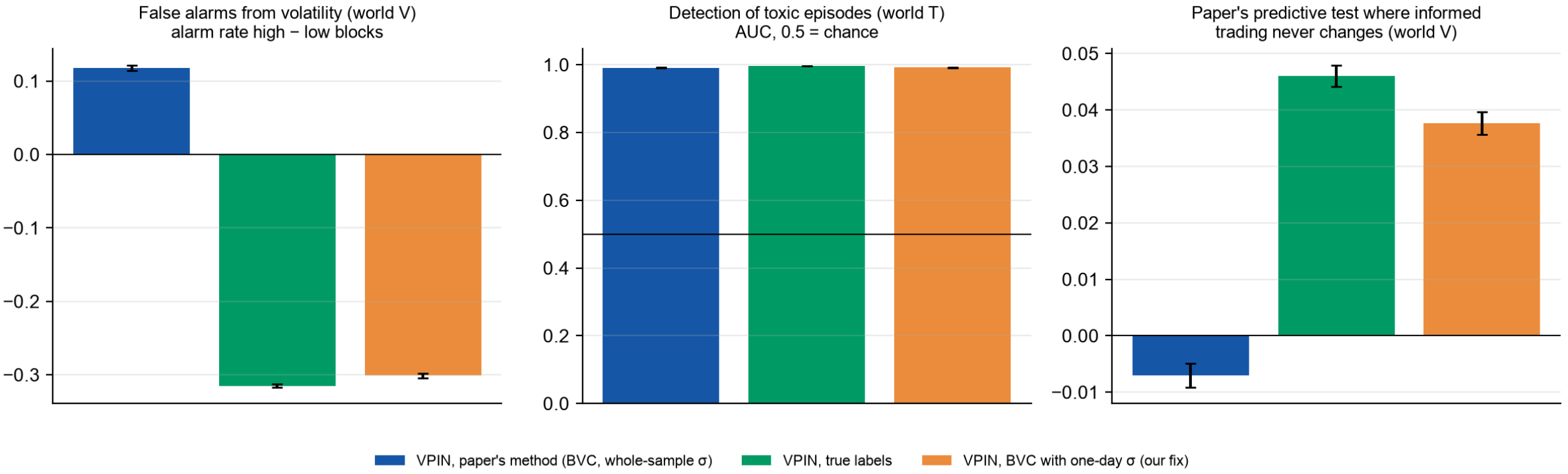
Market	Day	Largest 1-hour fall	VPIN alarm	VPIN alarm, true labels	Volatility alarm
BTC	2021-05-19	-22.9%	on	off	off
BTC	2024-08-05	-8.9%	off	off	off
BTC	2025-10-10	-11.6%	off	on	off
IVE	2011-08-08	-3.2%	on	–	on
IVE	2015-08-24	-5.1%	on	–	on
IVE	2018-02-05	-2.7%	on	–	off
IVE	2020-03-16	-4.6%	on	–	on
IVE	2024-08-05	-1.0%	on	–	off

	Alarm	Hits	On overall	p
BTC	VPIN	1 of 3	10%	0.261
BTC	VPIN, true labels	1 of 3	14%	0.355
BTC	Volatility	0 of 3	3%	1.000
IVE	VPIN	5 of 5	15%	< 0.001
IVE	Volatility	3 of 5	9%	0.006

• p: chance of that many hits if the alarm were on at random with its overall rate (falls treated as independent).

• IVE: VPIN on at every fall. BTC: nothing reliable. Few events.

Backup: the known-answer test, every metric



Blue: the paper's method (BVC, whole-sample σ); green: true labels; orange: BVC with one-day σ (our fix). Error bars: ± 1.96 Monte Carlo standard errors. Left: false alarms from volatility (world V). Middle: detection of toxic episodes (world T, AUC). Right: the paper's predictive test where informed trading never changes (world V).

Backup: repository and how to run it

```
src/vpin/
  data.py          IVE ticks, BTC klines and trades; cancelled trades
  bars.py          time bars, within-day price changes
  classify.py      BVC, tick rule, Lee-Ready
  buckets.py       equal-volume buckets
  metric.py        VPIN, full-sample and real-time CDF
  analysis.py      Tables 1–5, Figures 10–14
  realtime.py      no-look-ahead pipeline
  paper_numbers.py  the paper's numbers with pages
scripts/  download, reproduce, extensions, report, slides
tests/    unit tests (incl. Table 1 formula check)
DEVIATIONS.md  every deviation and silent decision
```

Run

pip install -r requirements.txt

pytest -q

python scripts/download_kibot.py

python scripts/download_binance.py

python scripts/reproduce.py

python scripts/extensions.py

python scripts/make_report.py

Free data: 942,321 IVE trades in the paper's period; BTCUSDT 2020–2026.